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Structure of the eye

Structures around the eyes

The eyes are situated in cavities in the skull called **orbits** (Fig. 12.1). These enclose and protect all but the front of the eyes.

Eyelids The exposed front of each eye is protected by a thin skin called the **conjunctiva**, and by the eyelids, eyelashes, and tear glands (Fig. 1). When the eyes are open, the eyelashes form a protective net which traps large particles of dust. The tear glands are situated above the eyeballs. They produce a continuous stream of watery liquid which is wiped over the conjunctiva every few seconds by automatic (reflex) blink movements of the eyelids. Tears keep the conjunctiva moist, wash it clean, and contain an enzyme which destroys bacteria. When dust or chemicals blow into the eyes, the flow of tears and blinking rate increase automatically until the eyes are cleaned.

Eye muscles Each eye is held in place and moved within its orbit by six **extrinsic muscles** attached to the outer surface of the eyeball and the skull. These muscles can rotate the eyes to follow moving objects, and direct the gaze to a chosen object. The movements are precisely co-ordinated so that both eyes work together and are always directed at the same spot.

Parts of the eyeball

Sclerotic layer This is a thick layer of tough white fibres which form the outer wall of the eyeball. The sclerotic protects the delicate inner structures and helps maintain the eye's rounded shape.

Cornea This is a transparent circular window at the front of the eyeball. It is continuous with the sclerotic. The curved shape of the cornea causes light rays to bend (be reflected) as they pass through it, so that they converge on the back of the eyeball.

Choroid layer This is a layer of blood vessels and cells full of dark brown pigment. It lines the inside of the sclerotic at the back of the eye. The blood vessels supply the eye with food and oxygen, and the pigment prevents light from being reflected around inside the eyeball. The front edge of the choroid forms a thick ring of tissue called the ciliary body.

Ciliary body This consists of blood vessels, and a mass of muscle fibres which make up the **ciliary muscle**. This muscle controls the shape of the lens during focusing, as described in Unit 47. Fibres called the **suspensory ligaments** run from the inner edge of the ciliary body to the outer rim of the lens. These fibres hold the lens in place and play a part in focusing.

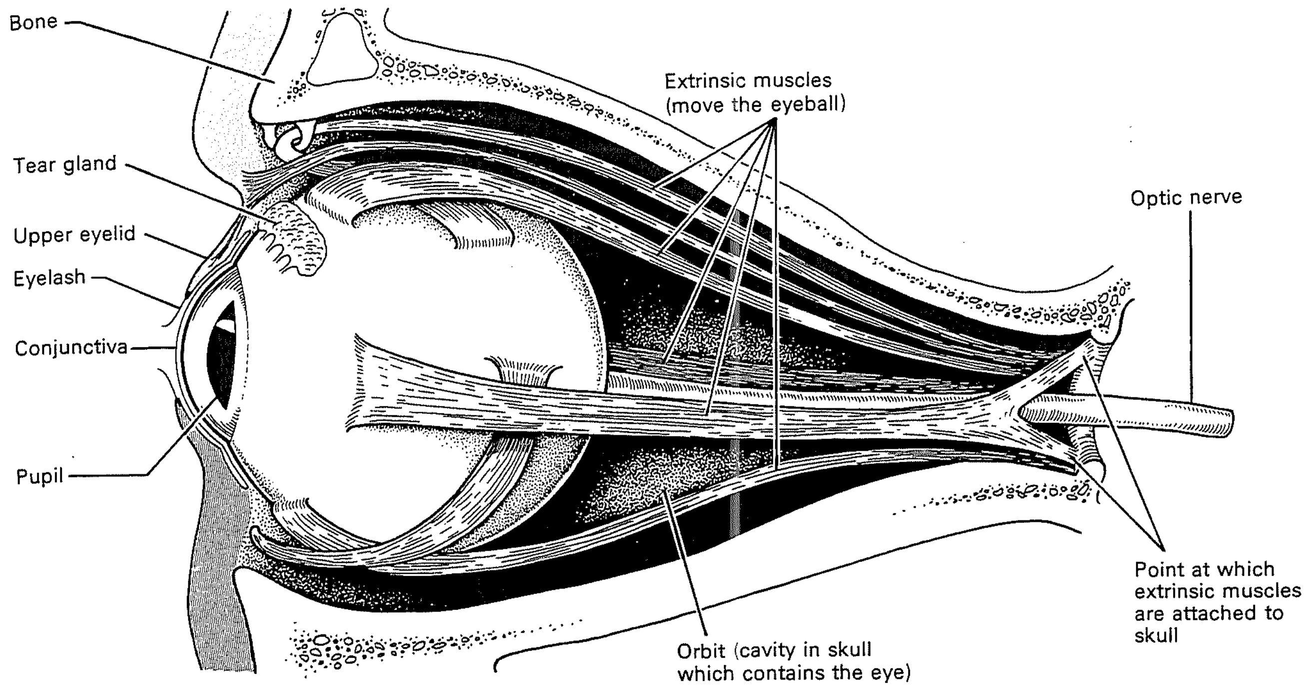
The lens This consists of layers of transparent tissue, arranged like the skins of an onion, which are enclosed in a transparent membrane. The lens is concerned with focusing a clear image on the back of the eye. The cornea alone can do this, but the lens makes it possible for the image to be refocused during shifts of vision between near and distant objects.

Iris This is the coloured part of the eye. It is situated in front of the lens and has a hole at its centre called the **pupil**. The iris is made up of radial and circular muscles. When the radial muscles contract the pupil is enlarged, and when the circular muscles contract the pupil is reduced in size. In this way the iris regulates the amount of light entering the eye: it opens the pupil in dim light and reduces it to pin-hole size in bright light.

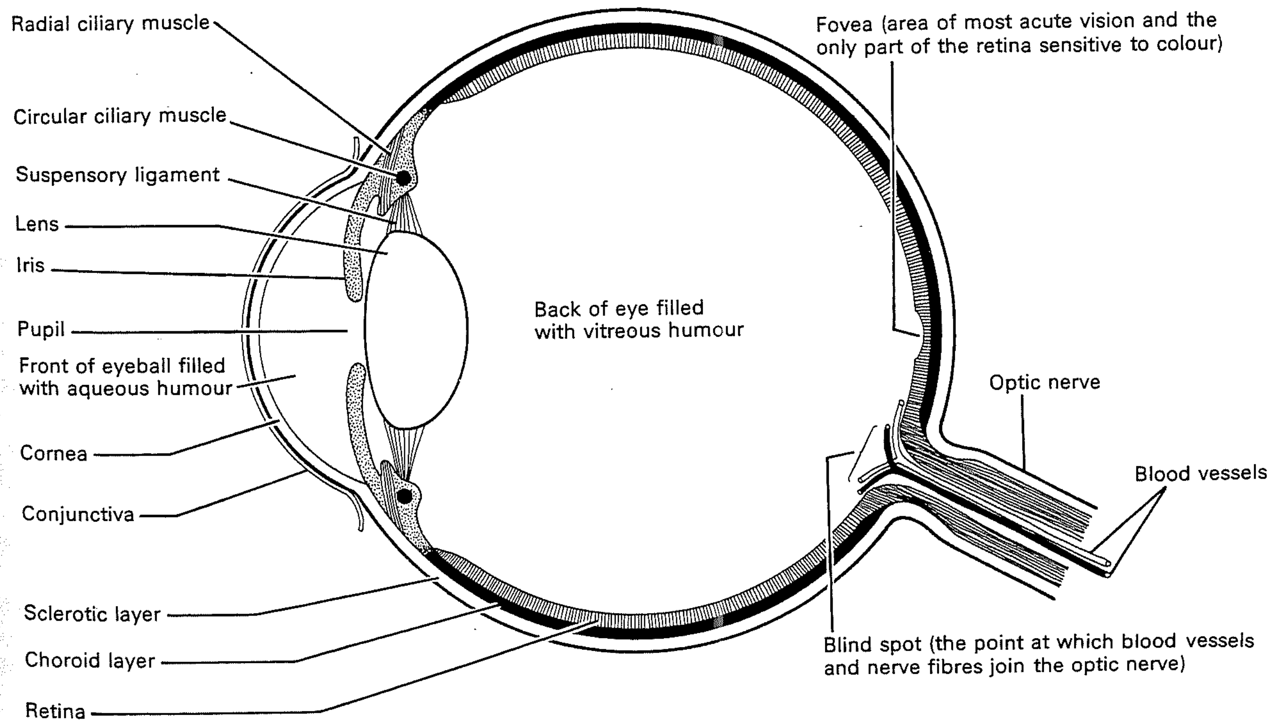
Retina This is a layer of sensory nerve endings which cover the back of the eye. The cornea and lens focus light rays on the retina forming a clear, upside-down, full-colour image of the outside world. The nerve endings in the retina are sensitive to light and respond to this image by sending nerve impulses along nerve fibres which extend from the eye and along the optic nerve to the brain. The retina and image formation are described in Unit 47.

Aqueous and vitreous humours The space in front of the lens is filled with a liquid called aqueous humour, and the space behind the lens is filled with a jelly called vitreous humour. Aqueous humour is similar to blood plasma. It is produced by blood vessels in the ciliary body, and carries food and oxygen from this blood supply to the cornea and lens. It also inflates the front of the eye, thereby maintaining the curved shape of the cornea. Aqueous humour drains out of the eye into the bloodstream through a tiny canal in the angle between the iris and cornea.

Vitreous humour gives shape and firmness to the rear compartment of the eye, and keeps the retina in contact with the choroid and sclerotic layers. Both the aqueous and vitreous humours play a part in focusing an image on the retina by bending light rays passing through them.



;. 1. **Structures attached to the eyeball.** Eyes are ›tected by sockets in the skull called orbits. The front of each : is protected by two eyelids, eyelashes, and a tear gland. Tears keep the eyes moist, wash away dirt, and contain an enzyme which destroys bacteria. Eyes are held in place and moved inside their sockets by extrinsic muscles.



ig. 2. **Horizontal section through the right eye.**

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Focusing and structure of the retina

Focusing (accommodation)

The technical name for refocusing the eye between near and distant objects is accommodation. This is accomplished by changing the shape of the lens, and is possible because the lens is made of elastic material. The shape of the lens is changed by contraction of the radial and circular ciliary muscles.

Accommodation for a distant object An eye is focused on a distant object by changing the lens to a flattened (less convex) shape. This reduces to a minimum the power of the lens to bend (refract) light. A flattened shape is needed to focus the almost parallel light rays from distant objects (Fig. 1A). The lens is flattened by contraction of the **radial ciliary muscles**. These pull against the **suspensory ligaments**, which pull against the lens stretching it into a flatter shape.

Accommodation for a near object An eye is focused on a near object by changing the lens to a rounded (more convex) shape. This increases the lens's power to refract light, which is necessary in order to focus the diverging light rays from a near object onto the retina (Fig. 1B). The lens is made more rounded by contraction of the **circular ciliary muscles** and relaxation of the radial ones. Circular ciliary muscles contract to form a circle with a smaller diameter. This reduces tension on the suspensory ligaments and allows the lens to become rounded in shape.

When a lens system is focused on a near object, its depth of focus is low. This means that a narrow zone, perhaps only a few millimetres wide, is clearly focused. However, when the eyes focus on a near object there is simultaneous contraction of the pupils, which increases the depth of focus.

The retina

The retina contains light-sensitive receptors of two types: about 125 million are called **rods**, and about 6 million are called **cones**, because of their shape.

Rods and cones do not face the light. Over most of the retina they are buried first under a layer of the nerve fibres which conduct impulses from the retina to the brain, and second under a layer of capillaries (Fig. 2B). But this is not true of an area of retina called the **fovea**. The fovea is a small depression in the retina immediately opposite the lens. There are no capillaries over this region and the layer of nerve fibres is very thin (Fig. 2A).

Rods and cones are not evenly distributed over the retina. The fovea consists exclusively of cones, packed tightly together with 147 million/mm^2. Elsewhere the retina consists mostly of rods.

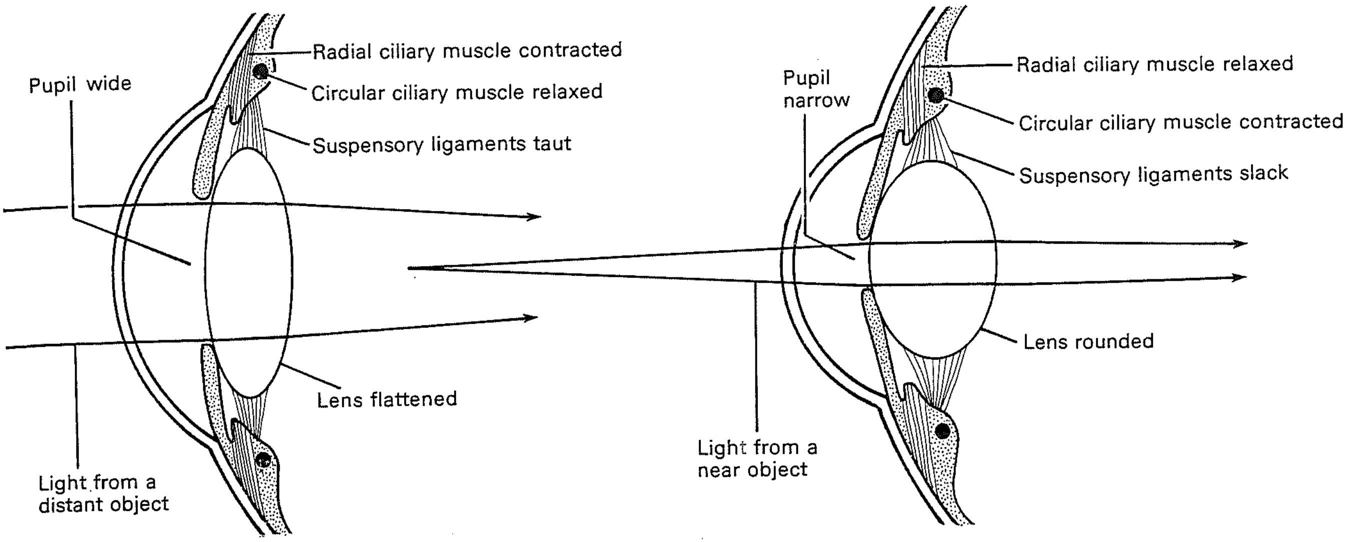
Groups of up to 150 rods are connected to the brain by only one nerve fibre, whereas each cone has either its own exclusive nerve fibre connection to the brain or it shares one with only a few others.

This means that an image falling on the fovea, where there are only cones, produces a much clearer visual impression in the brain than an image falling on the rest of the retina. This is because the part of an image which falls on the fovea is minutely analysed by its tightly packed cones, each of which sends separate impulses to the brain. In comparison, the image which falls outside the fovea is analysed by large patches of rods which share a single fibre connection to the brain. When someone wishes to examine something carefully, they automatically move their eyes or the object until its image falls on the fovea.

Rods at the outermost edge of the retina do not produce a visual impression in the brain. They serve only to trigger reflexes which turn the eyes towards objects just beyond the limits of normal vision. This is what happens when something is seen 'out of the corner of the eye'.

There are several other important differences between rods and cones. Rods continue to work in dim light, but they are not sensitive to colours. Cones, on the other hand, work only in bright light and are sensitive to colours. These facts explain the differences between day and night vision. Daylight vision is in colour, and has precise detail because it results mostly from images which fall on the cones of the fovea. However, the cones stop working as daylight fades, and vision relies more and more on the rods. Towards evening, a person loses the ability to see colours, and vision becomes less distinct.

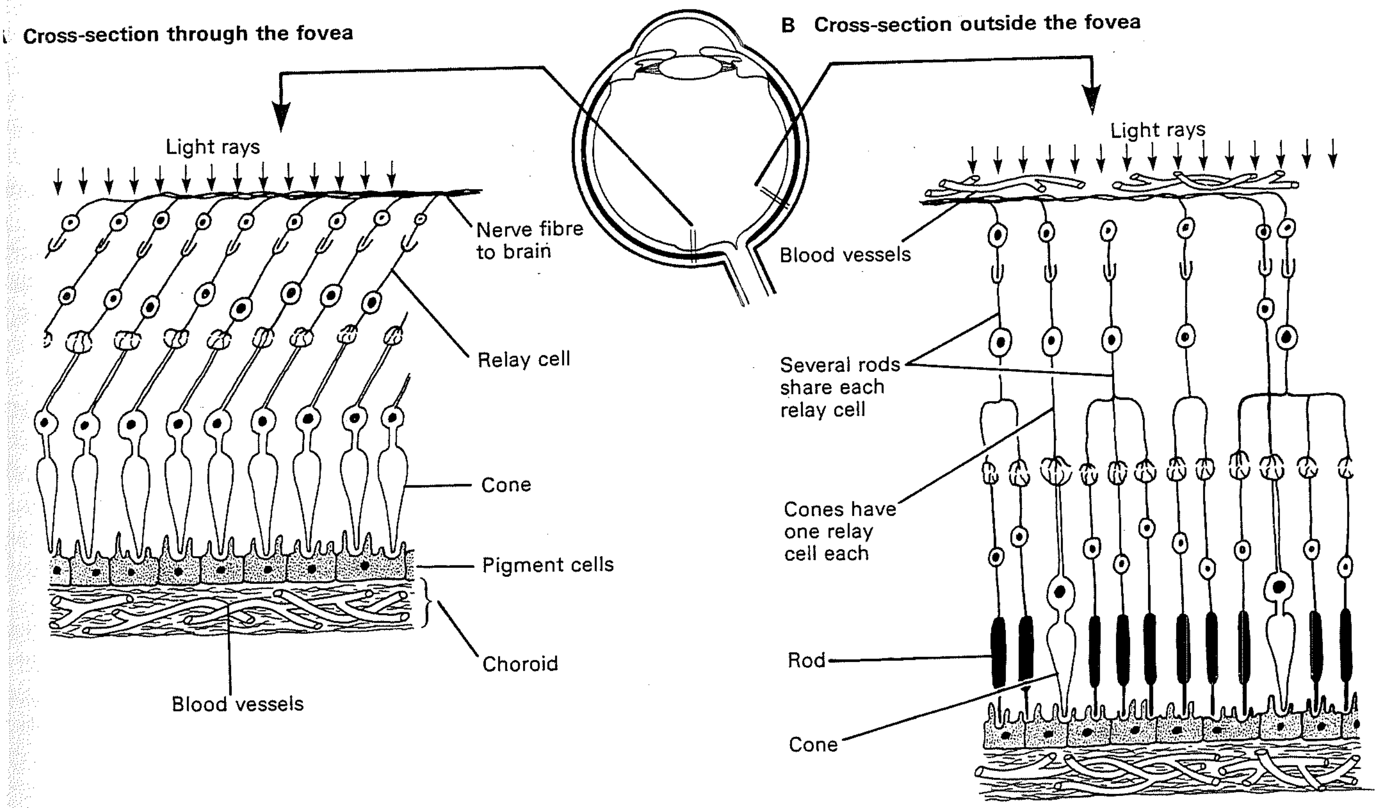
The **blind spot** is a point in the retina where blood vessels and nerve fibres leave the eye and form the **optic nerve** (Fig. 46.2). The blind spot contains neither rods nor cones therefore, as its name implies, it is entirely insensitive to light. The blind spot is described and demonstrated in Unit 48.



A Eye focused on a distant object
The radial ciliary muscles contract and pull against the suspensory ligaments. This stretches the lens into a flattened (less convex) shape

B Eye focused on a near object
Circular ciliary muscles contract. This releases tension on the suspensory ligaments and the lens becomes more rounded in shape

ig. 1. **Diagram of changes in the eye during focusing.**



ig. 2. **Structure of the retina.** It contains light-sensitive eceptors called rods and cones. Cones are concentrated in the fovea. They work only in bright light and are sensitive to colour. Rods work in dim light but are not sensitive to colour.

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The ear and hearing

Sound waves travelling through the air are collected by the funnel-shaped **pinna** of each ear (Fig. 2). The pinna directs sound waves down a short tube, the end of which is closed off by a sheet of skin and muscle called the **ear drum**. Sound waves make the ear drum vibrate in and out.

The structures described so far are part of the **outer ear**. Behind the ear drum is an air-filled space called the **middle ear**. A passage called the **Eustachian tube** connects this space with the back of the mouth. The Eustachian tube opens during swallowing, letting air in or out of the middle ear so that air pressure is always the same on both sides of the ear drum.

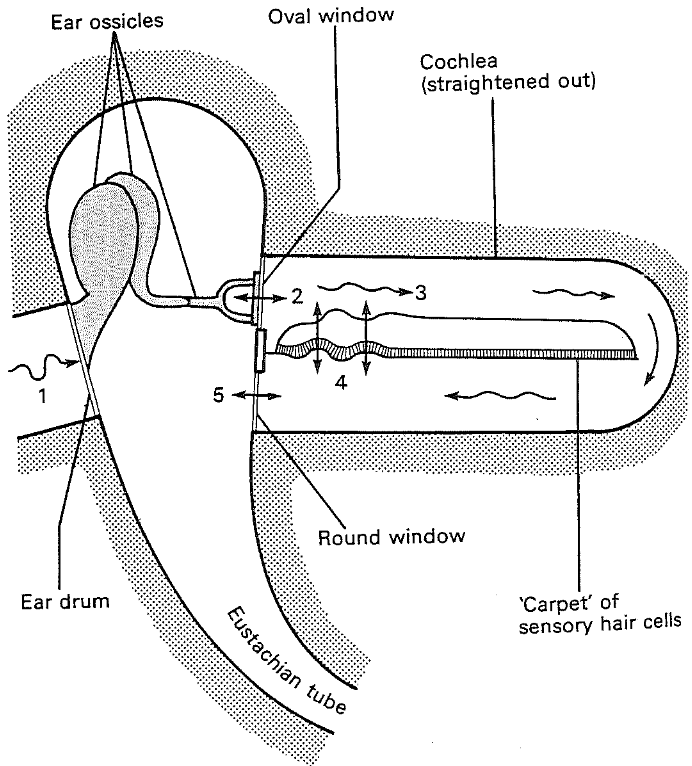
A chain of three tiny bones called the **ear ossicles** connects the ear drum with another sheet of skin called the **oval window**, which covers a tiny hole in the bones of the skull opposite the ear drum. When the ear drum vibrates, the ear ossicles move against each other in such a way that they lever the oval window in and out. These movements cause vibrations to pass into the inner ear.

The **inner ear** is a complicated series of tubes in the bones of the skull. The part of the inner ear concerned with hearing is called the **cochlea**, and is coiled like the shell of a snail (Fig. 2 A and B).

The cochlea is actually three tubes in one. The two outer tubes contain a liquid called **perilymph** and the middle tube contains **endolymph** (Fig. 2C). The floor of the middle tube is called the **basilar membrane**. This membrane supports a layer of sensory nerve endings called **hair cells** which stand upright like the pile of a carpet. The tips of the hairs are embedded in another membrane which overhangs them like a shelf jutting out of a wall.

Movements of the oval window cause vibrations (pressure waves) in the perilymph of the cochlea. These vibrations spread into the endolymph and finally cause the 'carpet' of hair cells to move up and down. This movement stimulates them to send nerve impulses along the auditory nerve to the brain, where they produce the sensation of sound (Fig. 1).

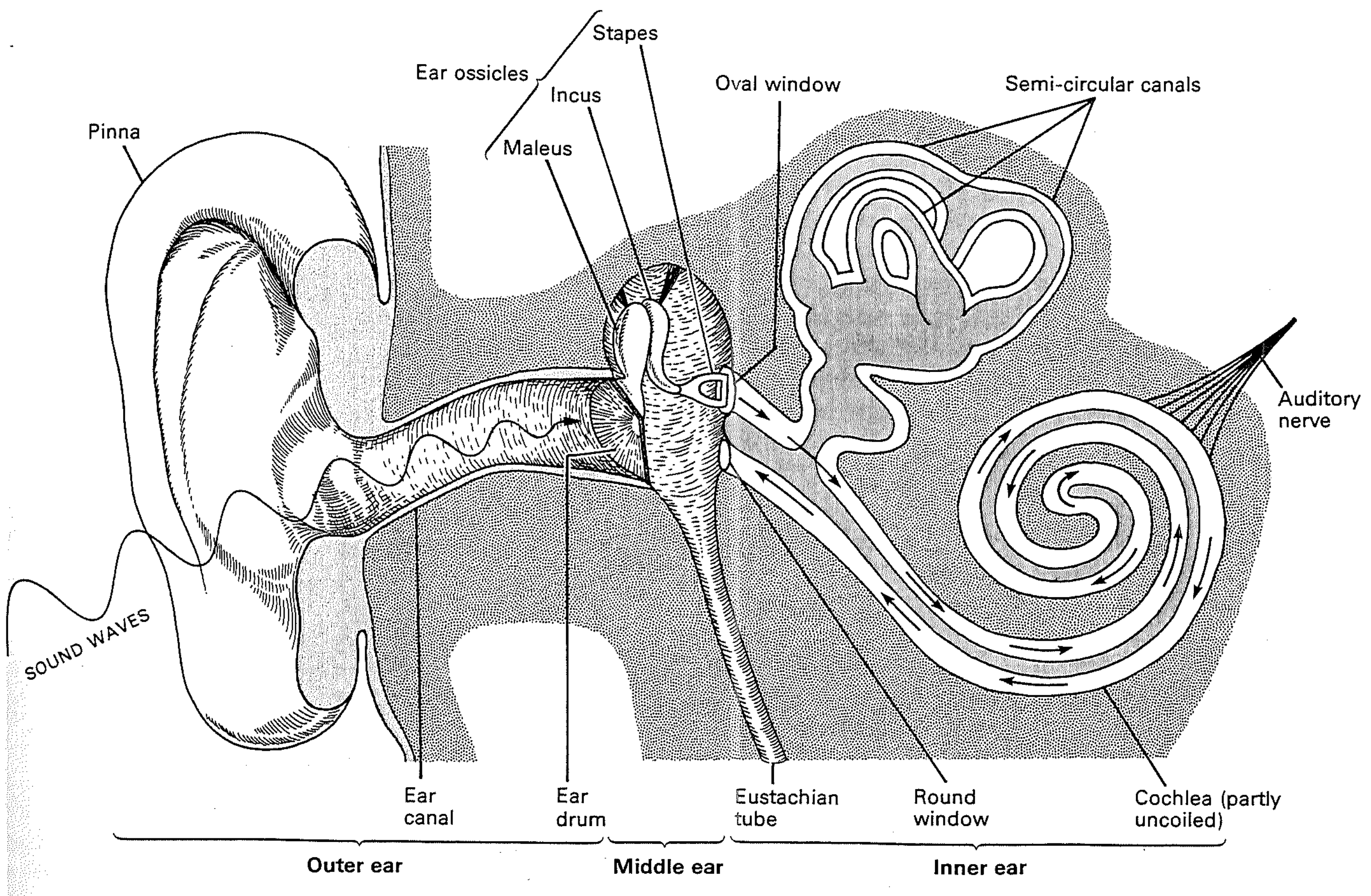
There is a theory that high-pitched notes cause vibrations only in hair cells nearest the oval window, while low-pitched notes cause vibrations further up the cochlea spiral. If this is true, the brain must distinguish notes from each other according to the region of hair cells stimulated as well as by the frequency of their vibrations. Whatever the mechanism it must be extremely sensitive, for humans not only distinguish between very similar notes at a wide range of different volumes, but between the same note played on many different instruments. The response of the ear to high-pitched sounds decreases with age and so old people gradually lose their ability to hear high notes.



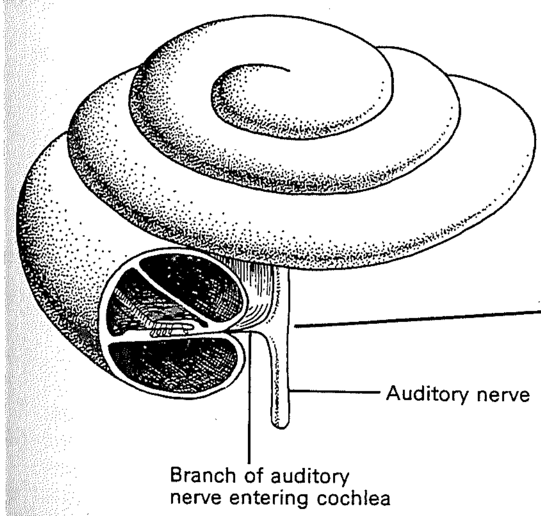
1. Sound waves in the air cause the ear drum to vibrate in and out
2. This causes the ossicles to vibrate and the stapes moves the oval window in and out
3. This sets up pressure waves in the cochlea
4. These pressure waves vibrate the sensory hair cells which send nerve impulses to the brain, causing the sensation of hearing
5. The pressure waves travel around the cochlea and move the round window in and out

Fig. 1. Diagram of how the ear changes sound waves into nerve impulses. (The cochlea is shown as a straight tube for the sake of simplicity.)

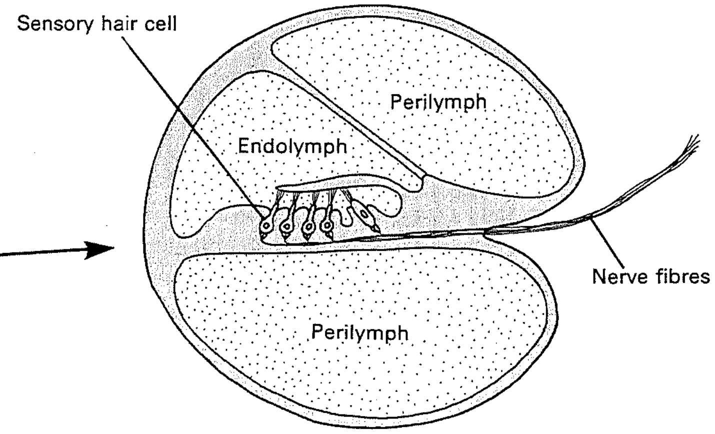
Structure of the ear (the middle and inner ears are diagrammatic and drawn on a larger scale than the outer ear)



Side view of cochlea (showing spiral shape)



C Cross-section through the cochlea



[illegible]-2. **Parts of the ear.**

7.8 Bionic ear and eye

Science as a human endeavour

FOUNDATION	STANDARD	ADVANCED

1. Use figures 7.8.1 and 7.8.2 to compare the human ear and a bionic ear; then complete the table below. Identify which parts of the human ear are equivalent (similar/the same) to the parts of the bionic ear. Where you do not think there is a direct equivalent, and why?

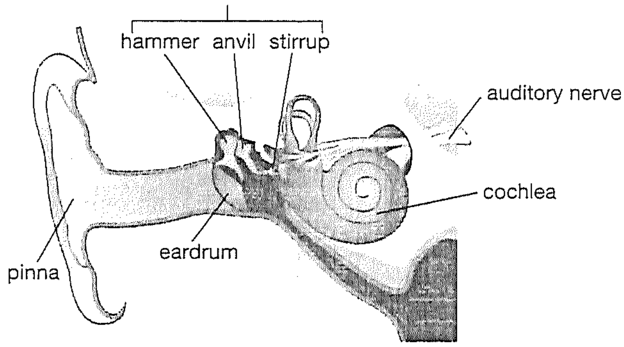


Figure 7.8.1 Structure of the human ear

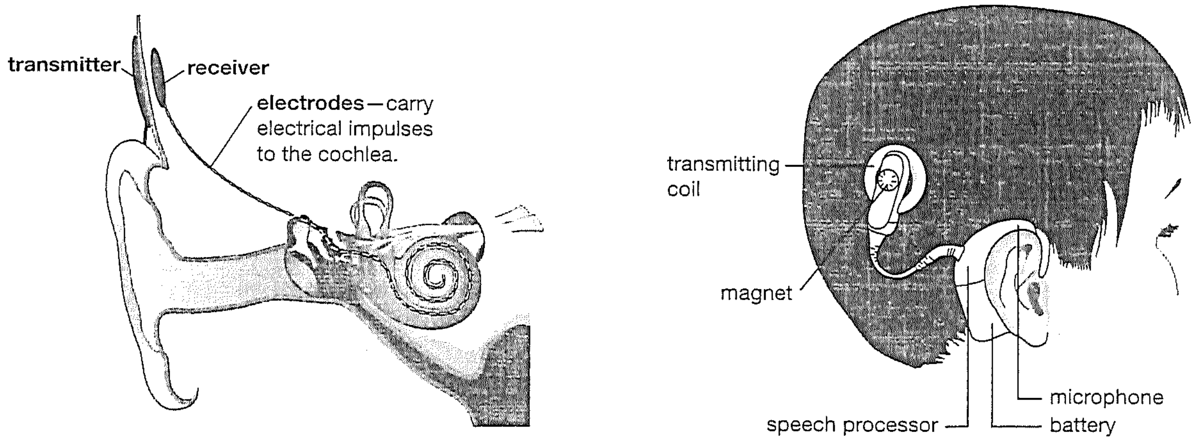


Figure 7.8.2 Structure of the bionic ear

Part of bionic ear	Equivalent in human ear	Justification
microphone		
speech processor		
transmitting coil		
receiver		
electrode		

(2) Although the bionic ear (Figure 7.8.2) and bionic eye (Figure 7.8.3) have different functions they have some things in common.

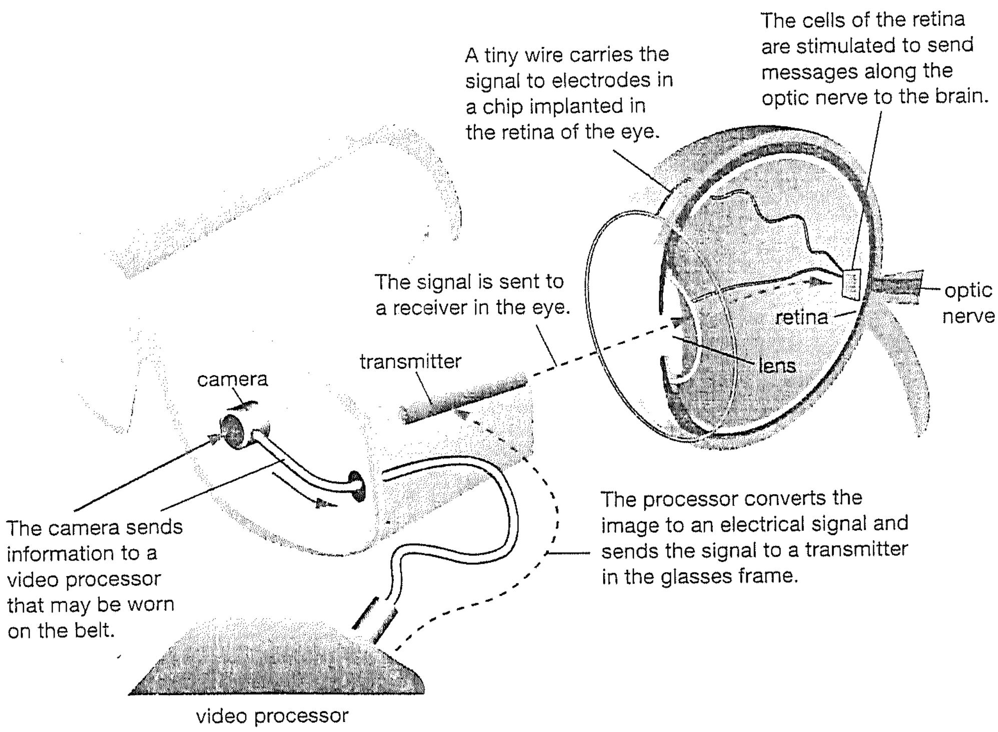


Figure 7.8.3 How a bionic eye works

Compare the bionic ear and the bionic eye.

3 What do you think are some of the short-term and long-term advantages of a young child being fitted with a bionic ear? Complete the following table.

Short- and long-term advantages of a bionic ear	
Short-term advantages	**Long-term advantages**

4 What do you think are some of the short-term and long-term advantages for the patient in having a bionic eye? Complete the following table.

Short- and long-term advantages of a bionic eye	
Short-term advantages	**Long-term advantages**

RATE MY UNDERSTANDING
Shade the face that shows your rating